

Challenge 3: Ramsey Numbers

Definition 1 (Ramsey Number). Given $t \in \mathbb{N}$, the (*diagonal*) *Ramsey number* $R(t)$ is the smallest $n \in \mathbb{N} \cup \{\infty\}$ such that every simple graph G on n vertices contains either a clique of size t or an independent set of size t .

By Ramsey's theorem, $R(t)$ is always an integer, which is upper bounded by 4^t . Erdős proved that random graphs $G(n, p)$ give a lower bound of the form $\sqrt{2}^t$ for t sufficiently large. An important problem that can be traced back to Ramsey's original paper is to get good upper and lower bounds for the Ramsey number.

Open Problem 1.1 (Scaled Ramsey problem). Let $r \in \mathbb{N}$. Provide decimal numbers d_1 and d_2 with $|d_2 - d_1| \leq (4 - \sqrt{2}) \cdot 0.9^r$ so that for all sufficiently large $t \in \mathbb{N}$

$$d_1^t \leq R(t) \leq d_2^t.$$

The scaled Ramsey problem for $r = 1$ follows by combining the above mentioned works of Ramsey and Erdős. The problem was solved for $r = 2$ by Campos, Griffiths, Morris and Sahasrabudhe combined with the better bound from the follow-up paper by Gupta, Ndiaye, Norin and Wei.